translationBridgeManagement

Service Description

Service ID: *"translationBridgeManagement"*

Abstract
This document describes the `translationBridgeManagement` service, version 1.0.0, produced by the TranslationManager support system of Arrowhead 5.2.2.

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1 Overview

This document describes the `translationBridgeManagement` service, which provides management of translation bridges on behalf of consumers: discovery, negotiation, abort and query. It is produced by the TranslationManager system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interfaces are specified in the HTTP/TLS/JSON IDD [2] and the MQTT/TLS/JSON IDD [3].

1.1 Significant Prior Art

New in Arrowhead 5.

1.2 How This Service Is Meant to Be Used

DynamicServiceOrchestration uses `discovery` and `negotiation` on behalf of the consumer during orchestration; operators query and abort bridges.

1.3 Important Delimitations and Dependencies

Access follows the management policy of the TranslationManager.

2 Service Interface

This section describes the interfaces to the `translationBridgeManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.

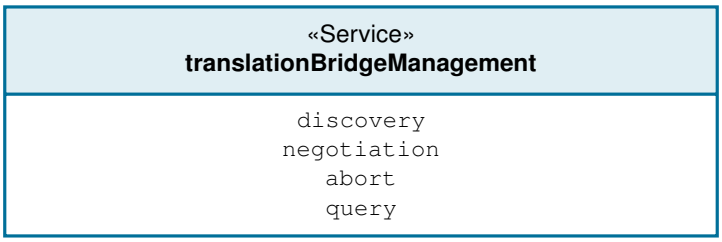


Figure 1: SysML-style block of the `translationBridgeManagement` service and its operations.

The following interface operations are available.

2.1 operation discovery (TranslationDiscoveryMgmtRequest) : TranslationDiscoveryResponse

Discovers translation candidates on behalf of the named consumer, with discovery flags.

2.2 operation negotiation (TranslationNegotiationMgmtRequest) : TranslationNegotiationResponse

Builds a discovered bridge to the chosen target instance on behalf of its consumer.

2.3 operation abort (List<Identifier>) : TranslationAbortMgmtResponse

Aborts the listed bridges; returns the result per bridge.

2.4 operation query (TranslationQueryRequest) : TranslationQueryListResponse

Returns a page of bridges matching the filter.

3 Information Model

Here, all data objects that can be part of the `translationBridgeManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.19, which are used to represent things like hashes and identifiers.

3.1 struct Address

A network address with its type.

Field	Type	Description
type	String	Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
address	String	One network address (host name, IPv4, IPv6 or MAC address).

3.2 struct DeviceResponse

A registered device.

Field	Type	Description
name	Name	Device name.
metadata	Metadata	Free-form metadata (JSON object).
addresses	List<Address>	Network addresses with their types.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.3 struct Page

Paging and sorting; all fields optional.

Field	Type	Description
page	Number	Zero-based page number.
size	Number	Number of records per page (at most `max.page.size`, default 1000).
direction	String	Sort direction, `ASC` or `DESC`.
sortField	String	Field to sort by.

3.4 struct ServiceDefinitionResponse

A service definition.

Field	Type	Description
name	Name	Service definition name.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.5 struct ServiceInstanceInterfaceResponse

One interface of a service instance.

Field	Type	Description
templateName	Name	Name of the interface template, snake_case (e.g. `generic_https`).
protocol	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
policy	String	Security policy of the interface: `NONE`, `CERT_AUTH` or one of the token policies.
properties	Map<String, Object>	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

3.6 struct ServiceInstanceResponse

A registered service instance.

Field	Type	Description
instanceId	Identifier	Service instance identifier `<System>|<serviceDefinition>|<version>`, or policy instance identifier.
provider	SystemResponse	The provider system.
serviceDefinition	ServiceDefinitionResponse	The service definition.
version	Version	Version of the service instance.
expiresAt	DateTime	UTC expiry time; no value means no expiry.
metadata	Metadata	Free-form metadata (JSON object).
interfaces	List<ServiceInstanceInterface Response>	Service interfaces offered by the instance.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.7 struct SystemResponse

A registered system.

Field	Type	Description
name	Name	System name.
metadata	Metadata	Free-form metadata (JSON object).
version	Version	Version of the system.
addresses	List<Address>	Network addresses with their types.
device	DeviceResponse	The device hosting the system, if registered.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.8 struct TranslationAbortMgmtResponse

Results of aborting bridges.

Field	Type	Description
results	Map<String, Boolean>	Per bridge identifier, whether it was aborted.

3.9 struct TranslationBridgeCandidate

A candidate reachable through translation.

Field	Type	Description
serviceInstanceId	Identifier	Service instance identifier `<System>|<serviceDefinition>|<version>`.
interfaceTemplateName	String	Interface template of the candidate the bridge would translate to.

3.10 struct TranslationDataModelTranslationDataDescriptor

Settings of a data-model translation.

Field	Type	Description
fromModelId	Identifier	Source data model.
toModelId	Identifier	Target data model.
interfaceProperties	Map<String, Object>	Interface properties of the translator.
configurationSettings	Map<String, Object>	Configuration passed to the translator.

3.11 struct TranslationDiscoveryMgmtRequest

Discovery of translation bridges on behalf of a consumer.

Field	Type	Description
candidates	List<ServiceInstanceResponse>	Candidate service instances (from service discovery).
consumer	Name	Consumer on whose behalf the bridge is discovered.
operation	String	QoS operation: `FILTER` or `SORT`.
interfaceTemplateNames	List<String>	Accepted interface templates, e.g. `generic_https`.
inputDataModelId	Identifier	Data model of the operation input expected by the consumer.
outputDataModelId	Identifier	Data model of the operation output expected by the consumer.
flags	Map<String, Boolean>	Discovery flags, e.g. `CANDIDATES_AUTH_CHECK`, `CONSUMER_BLACKLIST_CHECK`.

3.12 struct TranslationDiscoveryResponse

Candidates reachable through a translation bridge.

Field	Type	Description
bridgeId	Identifier	Identifier (UUID) of the translation bridge.
candidates	List<TranslationBridgeCandidate>	Store entries to create.

3.13 struct TranslationInterfaceTranslationDataDescriptor

Settings of an interface translation.

Field	Type	Description
fromInterfaceTemplate	String	Interface template of the consumer side.
toInterfaceTemplate	String	Interface template of the provider side.
token	Token	The token.
interfaceProperties	Map<String, Object>	Interface properties of the translator.
configurationSettings	Map<String, Object>	Configuration passed to the translator.

3.14 struct TranslationNegotiationMgmtRequest

Builds a discovered bridge on behalf of a consumer.

Field	Type	Description
bridgeId	Identifier	Identifier (UUID) of the translation bridge.
targetInstanceId	Identifier	Service instance chosen from the discovery candidates.

3.15 struct TranslationNegotiationResponse

The established bridge.

Field	Type	Description
bridgeId	Identifier	Identifier (UUID) of the translation bridge.
bridgeInterface	ServiceInstanceInterfaceResponse	Interface of the bridge the consumer calls instead of the provider.
tokenExpiresAt	DateTime	Expiry of the bridge token.
tokenUsageLimit	Number	Number of uses allowed with the bridge token.

3.16 struct TranslationQueryListResponse

List of translation bridges.

Field	Type	Description
entries	List<TranslationQueryResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

3.17 struct TranslationQueryRequest

Paged filter for translation bridges.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
bridgeIds	List<Identifier>	Bridge identifiers to match.
creators	List<String>	Systems that created the bridges.
statuses	List<String>	Bridge statuses to match.
consumers	List<Name>	Consumer systems to match.
providers	List<Name>	Provider system names to match.
serviceDefinitions	List<Name>	Service definition names to match.
interfaceTranslators	List<String>	Interface translator systems to match.
dataModelTranslators	List<String>	Data model translator systems to match.
creationFrom	DateTime	Lower bound of the creation time.
creationTo	DateTime	Upper bound of the creation time.
alivesFrom	DateTime	Lower bound of the bridge expiry.
alivesTo	DateTime	Upper bound of the bridge expiry.
minUsage	Number	Minimum number of usage reports.
maxUsage	Number	Maximum number of usage reports.

3.18 struct TranslationQueryResponse

A translation bridge.

Field	Type	Description
bridgeId	Identifier	Identifier (UUID) of the translation bridge.
status	String	Bridge status: `NEW`, `DISCOVERED`, `PENDING`, `INITIALIZED`, `USED`, `ABORTED`, `CLOSED` or `ERROR`.
usageReportCount	Number	Number of usage reports received for the bridge.
alivesAt	DateTime	Expiry of the bridge.
message	String	Text of the log entry or status message of the job.
consumer	Name	Name of the consumer system.
provider	Name	Name of the provider system.
serviceDefinition	Name	Service definition name.
operation	String	QoS operation: `FILTER` or `SORT`.
interfaceTranslator	String	System translating between the interfaces.
interfaceTranslatorData	TranslationInterfaceTranslationDataDescriptor	Settings of the interface translation.
inputDataModelTranslator	String	System translating the input data model.
inputDataModelTranslatorData	TranslationDataModelTranslationDataDescriptor	Settings of the input data-model translation.
outputDataModelTranslator	String	System translating the output data model.
outputDataModelTranslatorData	TranslationDataModelTranslationDataDescriptor	Settings of the output data-model translation.
createdBy	Name	Name of the system that created the record.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.19 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
Object	Any structured value.
String	A chain of characters.
Token	An opaque string used as credential.
Version	Specifies a version, semantic versioning `major.minor.patch`.

4 References

[1] J. Delsing, “TranslationManager – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-support-java-spring/TranslationManager/documentation/v5.2.2/TranslationManager-v5.2.2-SysD.pdf.

[2] ——, “translationBridgeManagement HTTP/TLS/JSON – Interface Design Description (TranslationManager),” Eclipse Arrowhead, IDD 5.2.2, 2026, translationManager-v5.2.2-translationBridgeManagement-HTTP-IDD.pdf.

[3] ——, “translationBridgeManagement MQTT/TLS/JSON – Interface Design Description (TranslationManager),” Eclipse Arrowhead, IDD 5.2.2, 2026, translationManager-v5.2.2-translationBridgeManagement-MQTT-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	